PRACTICAL 9. Hypothesis Testing

PRACTICAL NO.9(A)

Aim:- Formulate null and alternative hypothesis related to business problem, conduct a hypothesis test during appropriate statistical test, and interpret the result.

Code:-

```
from scipy.stats import ttest_1samp

#Sample delivery times
delivery_time=[28,31,29,32,30,27,33,31,29,30]

#Claimed average
claimed_mean=30

#Perform t-test
t_stat,p_value=ttest_1samp(
    delivery_time,
    claimed_mean
)

print("T-Statistic:",t_stat)
print("P-Value:",p_value)

#Significance level
alpha=0.05

if p_value<alpha:
    print("Reject the Null Hypothesis.")
    print("Averge delivery time is significantly different from 30 minutes.")
else:
    print("Fail to reject the null hypothesis.")
    print("There is not enough evidence to say that the average delivery time is different from
30 minutes.")
```

Output:-

T-Statistic: 0.0
P-Value: 1.0
Fail to reject the null hypothesis.
There is not enough evidence to say that the average delivery time is different from 30 minutes.

PRACTICAL NO.9(B)

Aim:- Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions.

Code:-

```
from statsmodels.stats.proportion import proportions_ztest

#Purchases
purchases=[10,15]

#Visitors
visitors=[100,100]

#Perform A/B test
z_stat,p_value=proportions_ztest(
    purchases,
    visitors
)

#Calculate conversion rates
rate_A=purchases[0]/visitors[0]*100
rate_B=purchases[1]/visitors[1]*100

print("Conversion Rate A:",rate_A,"%")
print("Conversion Rate B:",rate_B,"%")

print("Z-Statistics:",z_stat)
print("P-Value:",p_value)

#Decision
alpha=0.05

if p_value<alpha:
    print("Reject the Null Hypothesis.")
    print("There is a significant difference between A and B.")
else:
    print("Fail to reject the Null Hypothesis.")
    print("There is no significant difference between A and B.")
```

Output:-

Conversion Rate A: 10.0 %
Conversion Rate B: 15.0 %
Z-Statistics: -1.0690449676496971
P-Value: 0.28504940740261286
Fail to reject the Null Hypothesis.
There is no significant difference between A and B.